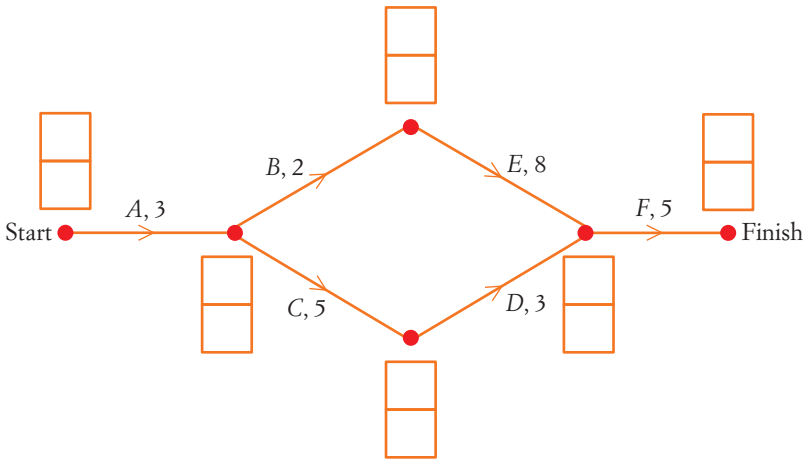
WORKSHEET

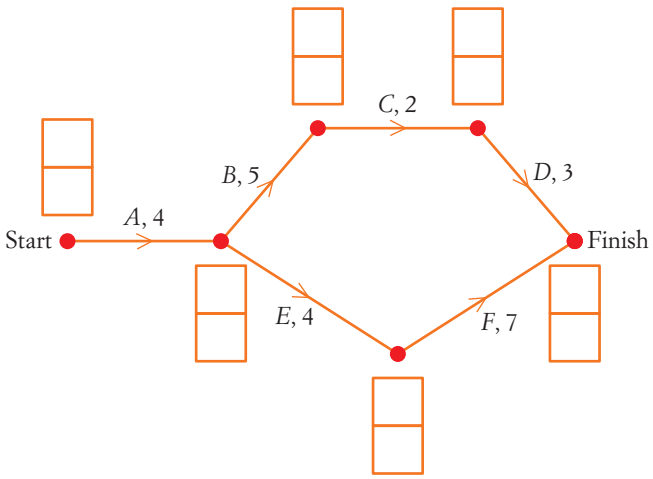
Chapter 9: Calculating EST and LST

1 Use forward and backward scanning to calculate the EST and LST of each of the following networks.

a

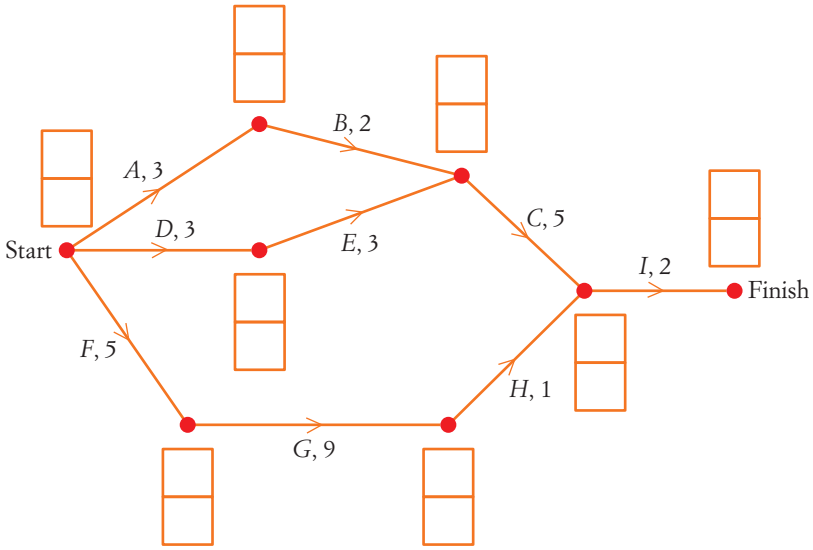


b

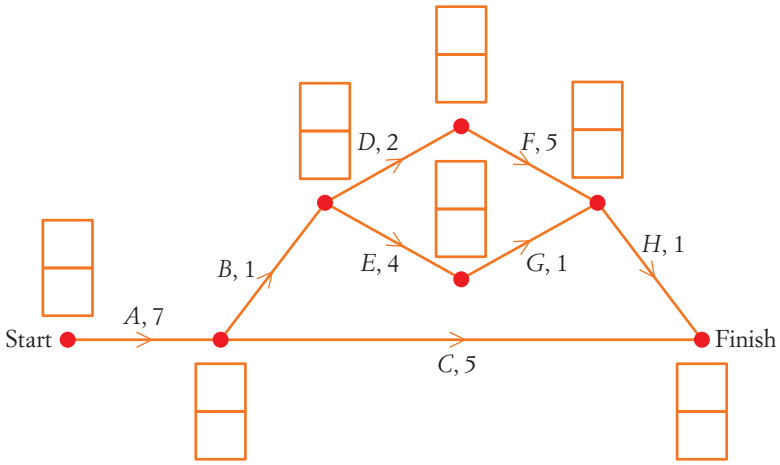


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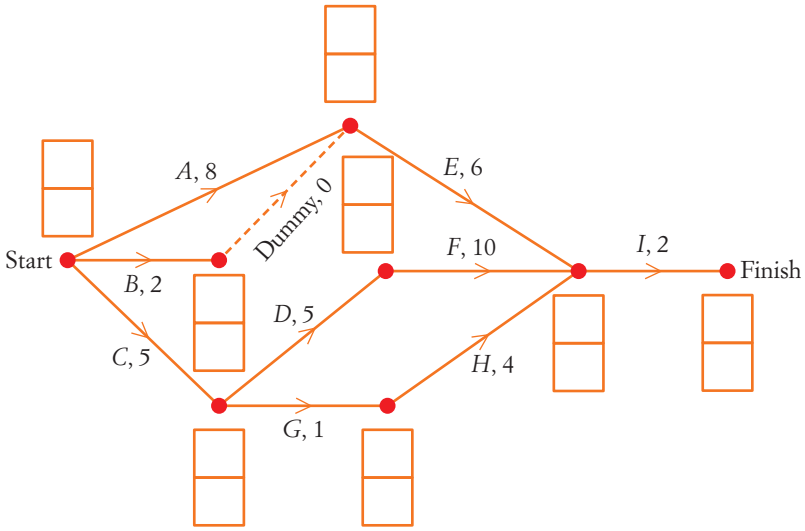
c



d



2 A directed graph is shown below.

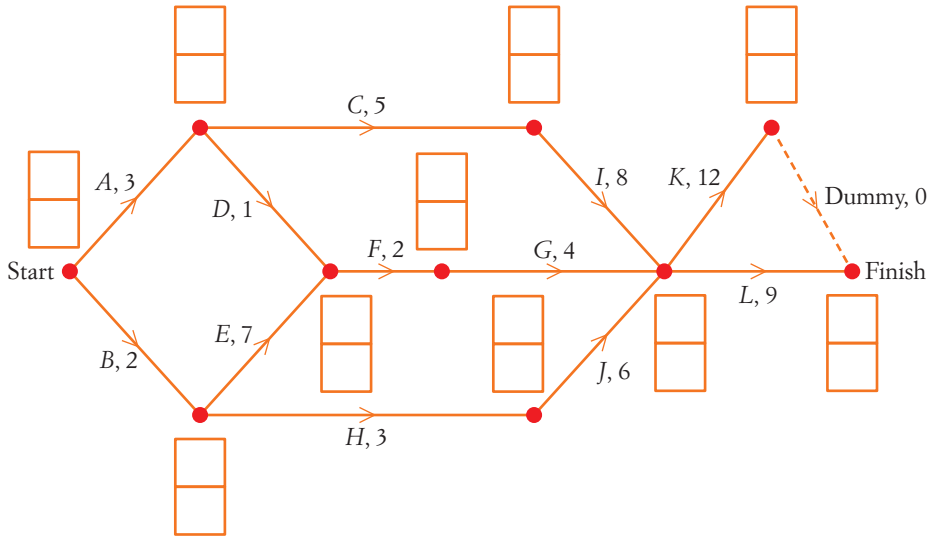


a Use forward and backward scanning on the directed graph to calculate the EST and LST of the network.

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b The time taken to complete activity *A* is reduced by 3 hours. What impact does this have on the EST and LST of the final vertex?

3 A directed graph is shown below.



a Use forward and backward scanning on the directed graph to calculate the EST and LST of the network.

b The time taken to complete activity *C* is increased by 4 hours. What impact does this have on the EST and LST of:

i the vertex following activities *I*, *G* and *J*

ii the final vertex?

4 The activity table for a project is shown below.

Activity	Time (days)	Immediate predecessor/s
A	5	None
B	1	*A*
C	1	*A*
D	2	*B*
E	3	*C*
F	4	*B*
G	2	*D*
H	5	*D*
I	7	*E*, *F*
J	2	*G*
K	3	*H*, *I*

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a Draw the network for this project.

b Use forward scanning to determine the EST for each activity.

c Use backward scanning to determine the LST for each activity.

5 The activity table for a project is shown below.

Activity	Time (days)	Immediate predecessor/s
A	9	None
B	12	*A*
C	3	*A*
D	2	*A*
E	3	*A*
F	4	*C*
G	7	*D*
H	2	*E*
I	5	*E*
J	6	*I*
K	2	*F*, *G*
L	6	*H*
M	8	*B*, *J*, *K*, *L*

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a Draw the network for this project.

b Use forward scanning to determine the EST for each activity.

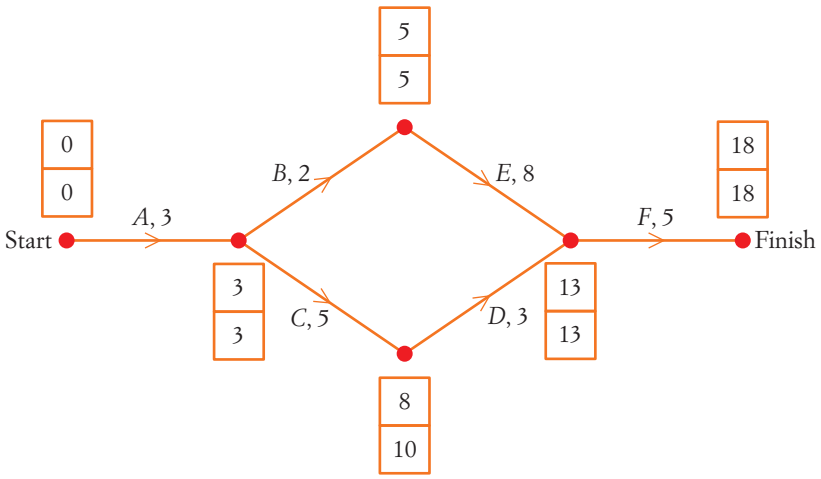
c Use backward scanning to determine the LST for each activity.

d The time taken to complete activity *J* is reduced by 4 hours. What impact does this have on the EST and LST of the final vertex?

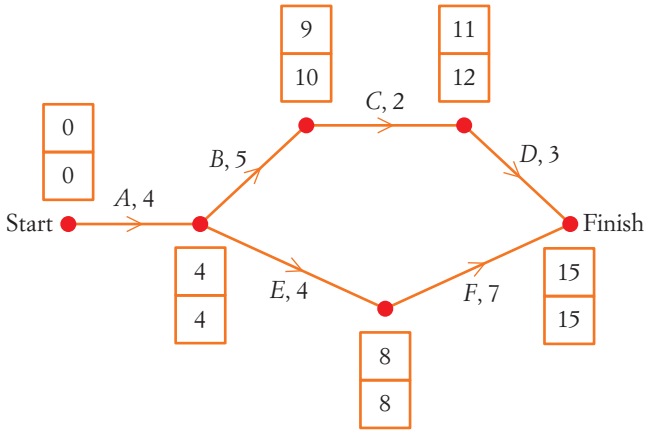
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Answers

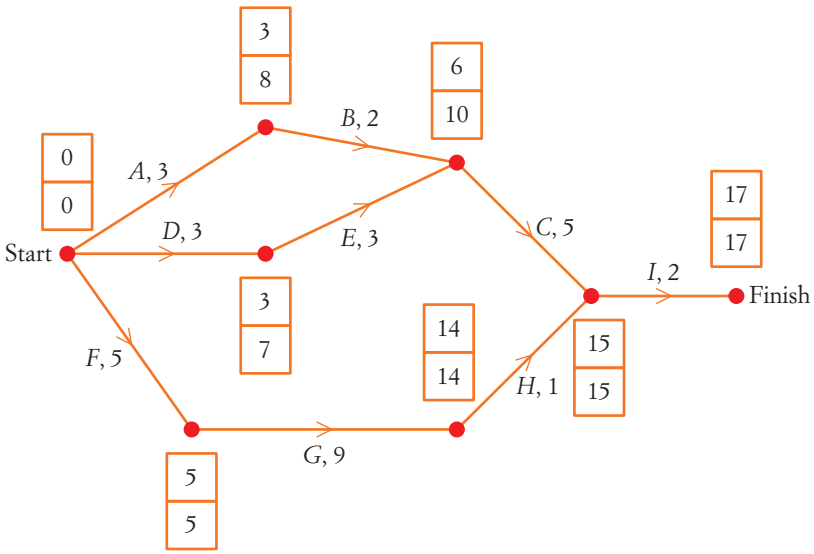
1 a



b

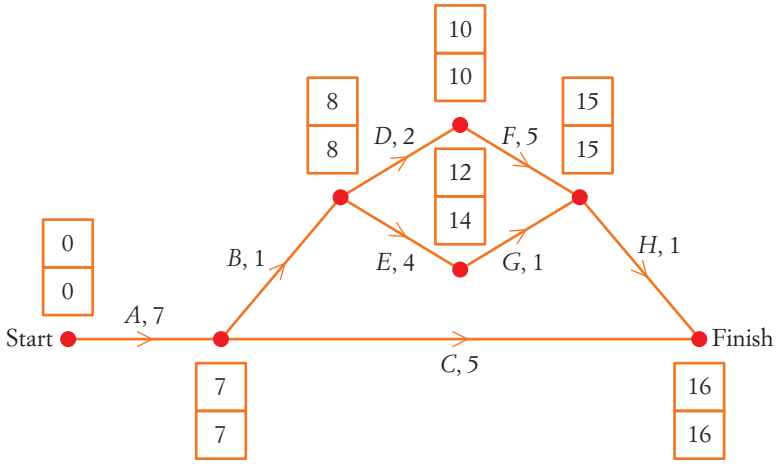


c

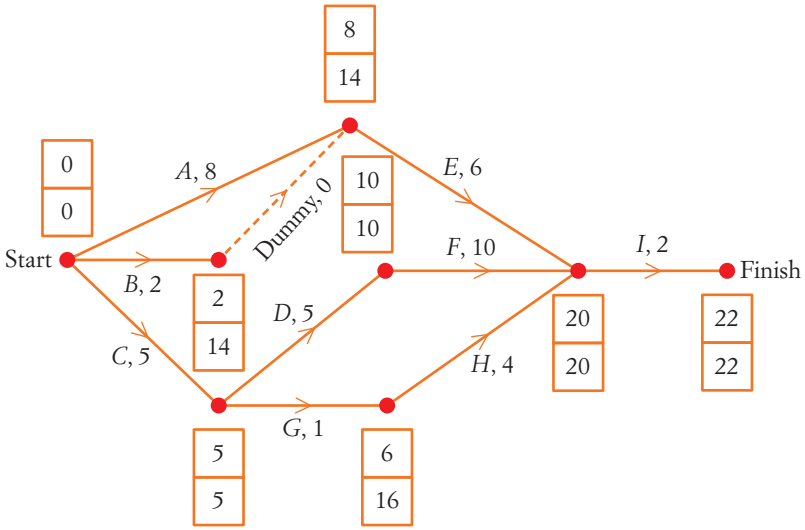


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d

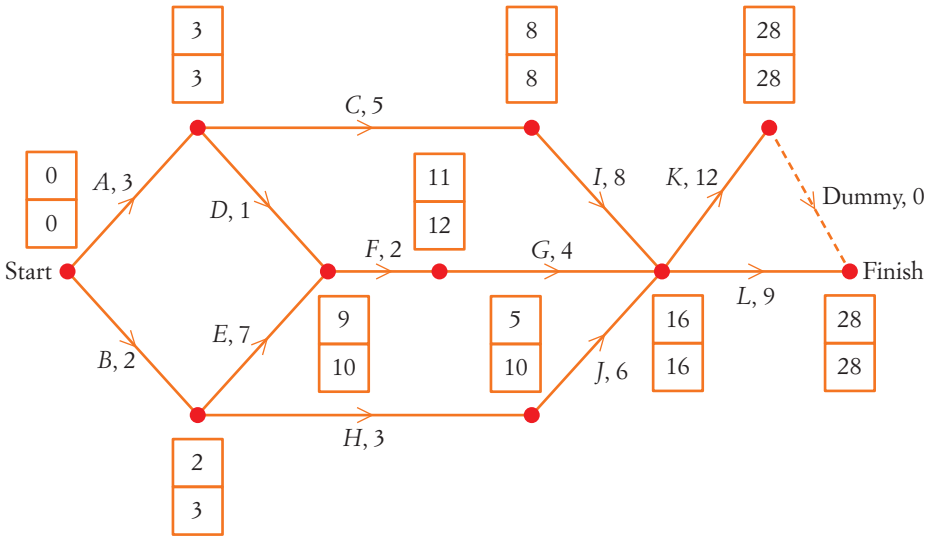


2 a



b No impact

3 a

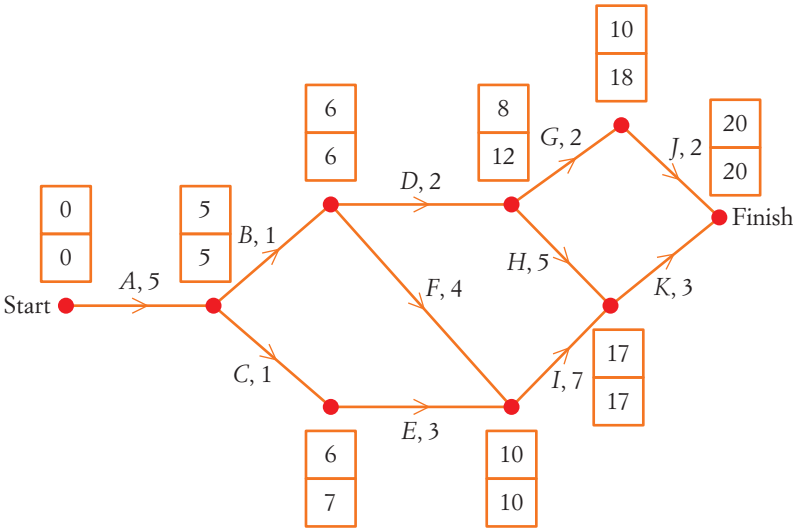


b **i** Increases both by 4 (to 20)

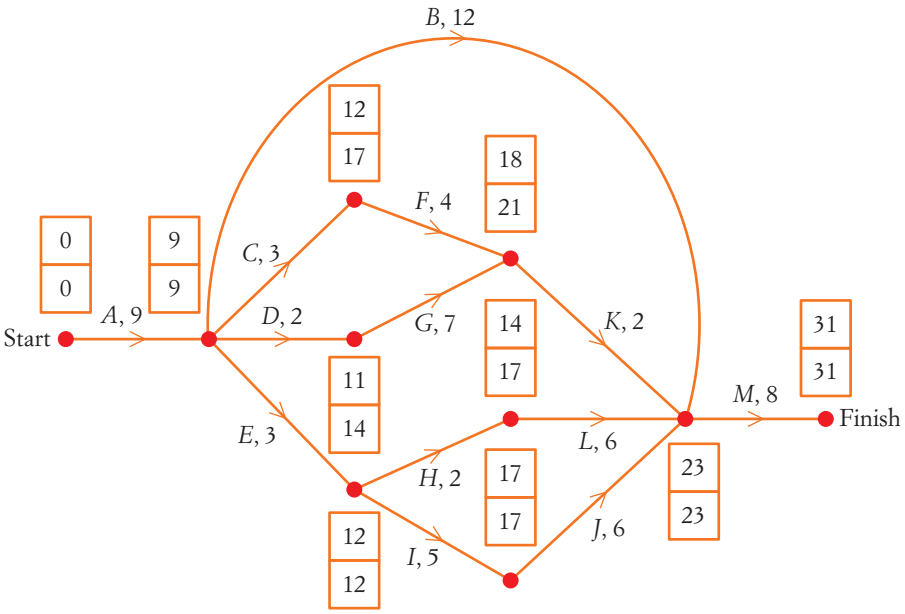
ii Increases both by 4 (to 32)

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4 a–c



5 a–c



d Reduces them by 3 hours

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